

Category: Reasoning

Question name: root_cause

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{'cvssMetricV31': {'Primary': {'source': 'nvd@nist.gov', 'type': 'Primary', 'exploitabilityScore':
2.8, 'impactScore': 5.9, 'version': '3.1', 'vectorString':
'CVSS:3.1/AV:N/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:H', 'baseScore': 8.8, 'baseSeverity': 'HIGH',
'attackVector': 'NETWORK', 'attackComplexity': 'LOW', 'privilegesRequired': 'NONE',
'userInteraction': 'REQUIRED', 'scope': 'UNCHANGED', 'confidentialityImpact': 'HIGH',
'integrityImpact': 'HIGH', 'availabilityImpact': 'HIGH'}}}, {'cvssMetricV2': {'Primary': {'source':
'nvd@nist.gov', 'type': 'Primary', 'baseSeverity': 'HIGH', 'exploitabilityScore': 8.6,
'impactScore': 10.0, 'acInsufInfo': False, 'obtainAllPrivilege': False, 'obtainUserPrivilege':
False, 'obtainOtherPrivilege': False, 'userInteractionRequired': True, 'version': '2.0',
'vectorString': 'AV:N/AC:M/Au:N/C:C/I:C/A:C', 'baseScore': 9.3, 'accessVector': 'NETWORK',
'accessComplexity': 'MEDIUM', 'authentication': 'NONE', 'confidentialityImpact': 'COMPLETE',
'integrityImpact':
impact scores
=====
Adobe Security Bulletin APSB16-25 for Adobe Flash Player describes multiple flaws that can possibly
lead to code execution when Flash Player is used to play a specially crafted SWF file. Quoting from
the APSB16-25: These updates resolve a race condition vulnerability that could lead to information
disclosure ( CVE-2016-4247 ). These updates resolve type confusion vulnerabilities that could lead
to code execution ( CVE-2016-4223 , CVE-2016-4224 , CVE-2016-4225 ). These updates resolve use-
after-free vulnerabilities that could lead to code execution ( CVE-2016-4173 , CVE-2016-4174 , CVE-
2016-4222 , CVE-2016-4226 , CVE-2016-4227 , CVE-2016-4228 , CVE-2016-4229 , CVE-2016-4230 , CVE-
2016-4231 , CVE-2016-4248 , CVE-2016-7020 ). These updates resolve a heap buffer overflow
vulnerability that could lead to code execution ( CVE-2016-4249 ). These updates resolve memory
corruption vulnerabilities that could lead to code execution ( CVE-2016-4172 , CVE-2016-4175 , CVE-
2016-4179 , CVE-2016-4180 ,
redhat-bugzilla
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This issue has been addressed in the following products: Red Hat Enterprise Linux 5 Supplementary
Red Hat Enterprise Linux 6 Supplementary Via RHSA-2016:1423 https://access.redhat.com/errata/RHSA-
2016:1423
redhat-bugzilla
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[{'id': 'cpe:2.3:a:adobe:flash_player_desktop_runtime:****:*:*:*:*:*', 'Vendor Name': 'adobe',
'Product Name': 'flash_player_desktop_runtime', 'Version': '*'}, {'id':
'cpe:2.3:a:adobe:flash_player:****:*:*:*:*:*', 'Vendor Name': 'adobe', 'Product Name':
'flash_player', 'Version': '*'}, {'id': 'cpe:2.3:a:adobe:flash_player:****:*:*:*:*:*:chrome:*:*',
'Vendor Name': 'adobe', 'Product Name': 'flash_player', 'Version': '*'}, {'id':
'cpe:2.3:a:adobe:flash_player:****:*:*:*:*:*:edge:*:*', 'Vendor Name': 'adobe', 'Product Name':
'flash_player', 'Version': '*'}, {'id':
'cpe:2.3:a:adobe:flash_player:****:*:*:*:*:*:internet_explorer:*:*', 'Vendor Name': 'adobe', 'Product
Name': 'flash_player', 'Version': '*'}, {'id': 'cpe:2.3:a:adobe:flash_player:****:*:*:*:*:*:*:*',
'Vendor Name': 'adobe', 'Product Name': 'flash_player', 'Version': '*'}]
impact scores
=====
'MEDIUM', 'authentication': 'NONE', 'confidentialityImpact': 'COMPLETE', 'integrityImpact':
'COMPLETE', 'availabilityImpact': 'COMPLETE'}}}
impact scores
=====
{'Ubuntu': {'cvss_score': '8.8', 'cvss_severity': 'High', 'ubuntu_priority': 'Medium'}}
```

Question: Explain the root cause of the vulnerability?

Answers

Adobe Flash Player contains multiple vulnerabilities that can lead to code execution.

Faithfulness: Is this claim strictly supported by the context?

Yes

No

Correctness: Is this claim factually correct as per the context?

Yes

No

The vulnerabilities in Adobe Flash Player include race condition vulnerabilities, type confusion vulnerabilities, use-after-free vulnerabilities, heap buffer overflow vulnerabilities, and memory corruption vulnerabilities.

Faithfulness: Is this claim strictly supported by the context?

Yes

No

Correctness: Is this claim factually correct as per the context?

Yes

No

The race condition vulnerability in Adobe Flash Player can lead to information disclosure.